

Preparation and characterization of Nd doped gadolinium gallium garnet nanopowders and crystals

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The structural and optical features of gallium gadolinium garnet (GGG) nanopowders doped with neodymium were investigated. Nanopowders of GGG:Nd were prepared by modified sol-gel method using acetic acid as complexing agent. This way permitted to incorporate large amounts of dopant (up to 10 %) without destroying garnet structure. Small single crystals of Nd:GGG were grown by a μ -pulling down method and spectroscopic features of nanopowders and their single crystal counterparts were compared. It has concluded that the Nd³⁺ ions are located preferentially in the same type of sites in crystal lattices of GGG:Nd nanopowders and their single crystal counterparts. In addition, it follows from the perfect agreement of emission wavelengths and line width recorded for nanopowders and single crystal samples that the crystal lattice of GGG in nanopowders is not distorted.

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1 Introduction

Gadolinium gallium garnet, Gd₃Ga₅O₁₂ (GGG) crystals are known as substrate for magnetic garnet bubble domain devices and laser materials. The GGG crystals doped by e.g. Nd are considered as promising for laser technology. GGG melts congruently at about 1800 °C. Crystals are grown from iridium crucible by Czochralski method and by μ -pulling down method.

The evaporation of gallium oxide from GGG and reaction with crucible material (iridium) at high temperatures complicates the process of crystallization and various defects are observed [1–3]. It was reported that the defects of GGG crystals depend on melt stoichiometry, crystal growth atmosphere as well as solid-liquid interface shape [3]. Brandle et al. [4] was the first who considered influence of oxygen on the GGG growth and the later thermodynamic calculations confirmed that 1% oxygen is the optimum content in the Czochralski growth atmosphere [1]. However higher oxygen's concentration 3–5 vol % also were applied [5,6]. In order to compensate the losses of evaporated gallium oxide during growth, sometimes excess of Ga₂O₃ is added into the starting material [7,8]. The losses of gallium oxide occur during crystals growth performed by μ -pulling down method [9], too. Using as starting material one phase polycrystalline GGG did not solve this problem [10]. Therefore, the synthesis at temperatures much lower than the melting point is reasonable. Wet chemical methods: co-precipitation [11,12], combustion [13,14] and modified sol-gel (Pechini) synthesis [15–18] were also used to produce nanopowders GGG. Unfortunately, it appears that the emission characteristics of luminescent ions in these materials depend crucially on the method of preparation [17]. In this work we present

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results of spectroscopic investigation of neodymium doped GGG nanocrystals obtained by modified sol-gel process which was successfully applied for preparing the nanopowder of yttrium aluminium garnet (YAG), yttrium oxide (YO) and calcium gadolinium oxyborates (CGB) [19-21]. In contrast to synthesis described in [15-17] the acetic acid was used as complexing agent instead of citric acid. It was proved by infrared spectroscopy FT-IR that COOH groups from acetic acid remain in xerogel during calcinations up to 850 °C. The chelate ligands protect the gentle non-organic matrix from break-up and distortion and make it possible to obtain nanocrystals with low content of structural defects. The drawback of using acetic acid is its high temperature of thermal decomposition and consequently high temperature of calcinations (at least 900 °C).

The results of the synthesis and characterization of GGG nanopowders pure and Nd-doped are presented. The maximum content of Nd in GGG - 10 at% was related to the expected phenomenon self-quenching of luminescence. To examine the influence of preparation procedure on characteristics of the GGG:Nd system, single crystals of Nd:GGG were grown by a μ -pulling down method and spectroscopic features of nanopowders and their single crystal counterparts were compared.

2 Experimental details

Sol-gel method Nanopowders of gadolinium gallium garnet pure and doped with several concentrations of neodymium: 0.1; 0.2; 1; 5; 10 at% were synthesized by modified sol-gel method. Gallium oxide was dissolved in nitric acid. The solvent was evaporated and deionized water was added. Gadolinium and neodymium oxides were dissolved in solution of acetic acid for several hours. The solutions were mixed together and stirred for 2 hours at 65 °C. Then ethylene glycol as a cross-linking agent was added and the solution was stirred again for 2 h. The obtained mixture was evaporated slowly, dried at 120 °C for 12 h and then grinded in an agate mortar to fine powder. These powders were calcined in air at 1100 °C for 12 h.

Crystal growth GGG crystals were grown in Nürmont GmbH apparatus by using the micro-pulling-down technique described in detail in [22]. Starting material were Nd-doped nanopowders prepared following the procedure described above as well as mixtures of the (high purity) metal oxides. The latter were mixed in a molar ratio $[\text{Gd}_2\text{O}_3]/[\text{Ga}_2\text{O}_3] = 38.0/62.0$ roughly corresponding to the congruently melting composition [23,24] and pre-reacted at 1500 °C for 48 hours. Completeness of the garnet formation reaction was verified by X-ray powder diffraction analysis. Nd was added to the undoped powder in form of very small amounts Nd_2O_3 increasing the rare earth (RE) oxide to gallium oxide ratio only marginally is still inside the range of congruent compositions found in literature.

The starting material was melted in an inductively heated iridium crucible thermally insulated by zirconia and alumina ceramics. To reduce evaporation from the superheated melt the crucible was enclosed on top with an iridium lid containing only a small degassing hole. All experiments were carried out in N_2 atmosphere containing 13 vol% CO_2 . The advantage of such gas mixture is an oxygen partial pressure increasing with temperature so that both, (i) oxidation of the iridium crucible at moderate, and (ii) reduction of Ga_2O_3 at elevated temperatures can be avoided efficiently [25]. A randomly oriented piece of GGG crystal was used as seed. Crystals were pulled with a rate of about 0.5 mm/min and the heating power adjusted to produce a crystal about 1 mm in diameter. Crystals which were detailed investigated (0.3 at% Nd^{3+} and 0.2 at% Nd^{3+}) are shown in figure 1 and their growth parameters are summarized in table 1.

Since GGG melt wets iridium well, diameter control was difficult when a flat-bottomed crucible was used [26]. Use of a crucible with a short nozzle improved the situation significantly and crystals with fairly constant diameter could be grown by adjustments of the heating power (Fig. 1).

Table 1 Parameters of crystal growth by micro-pulling-down method.

Crystal	Nd content [at%]	Material	Pulling rate [mm/min]	Remarks
GGG-03	0.30	oxides mixture	0.35	flat-bottomed crucible
GGG-05	0.20	nanopowder sol-gel	0.50 and 0.75	flat-bottomed crucible
GGG-06	0.20	nanopowder sol-gel	0.50 and 0.75	crucible with nozzle



Fig. 1 Crystals obtained by μ -pulling down method: 03) GGG: 0.3 at% Nd^{3+} from oxides; 05) GGG: 0.2 at% Nd^{3+} from nanopowder obtained by modified sol-gel method; 06) GGG: 0.2 at% Nd^{3+} from nanopowder obtained by modified sol-gel method by using crucible with nozzle.

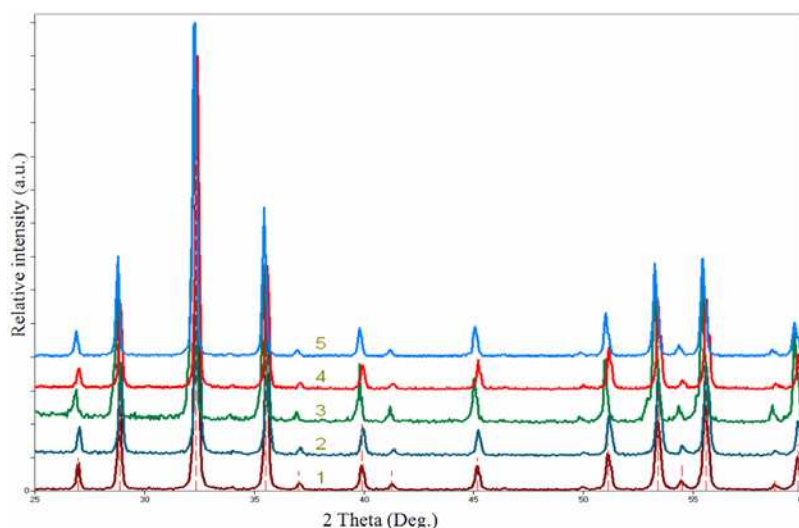


Fig. 2 X-ray diffraction patterns of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$: (1) undoped nanopowder; (2) nanopowder doped by 0.2 at% Nd^{3+} ; (3) crystal obtained from nanopowder doped by 0.2 at% Nd^{3+} ; (4) nanopowder doped by 1 at% Nd^{3+} ; (5) nanopowder doped by 10 at% Nd^{3+} ; nanopowders were heated at 1100 °C for 12 h. (Online color at www.crt-journal.org)

One can see that only crystals GGG-03 (0.3 at% Nd^{3+}) and GGG-06 (0.2 at% Nd^{3+}) were fully transparent. Their spectroscopic features were investigated. The crystal GGG-06 is considered as the best. Its diameter did not change during growth. This crystal was grown from nanopowder GGG: 0.2% Nd^{3+} . Also the crystal GGG-05 was grown from the same powder as GGG-06 but by using the crucible with different shape. For crystal GGG-05 was used flat-bottomed crucible and for crystal GGG-06 crucible with nozzle (table 1). The nozzle at the bottom of the crucible from which the crystal GGG-06 was grown plays the part of a shaper. It restricts the extension of the liquid zone and allows to reach the field of capillary stability distinguished by (nearly) constant diameter growth. For this growth regime, flow of matter and heat are stable in time.

3 Characterization of nanopowders and crystals

The crystal structures of samples were characterized by X-ray diffraction (XRD) using a Siemens D-500 diffractometer with CuK_α radiation at 1.548 Å. The size and morphology of nanopowders were analyzed by high resolution scanning electron microscopy (HRSEM) with LEO GEMINI 1530 apparatus. The spectra were excited at 806 nm by an AlGaAs laser diode, dispersed by a grating monochromator with 75 cm focal length and detected by a Hamamatsu R5108 photomultiplier.

X-ray diffraction method (XRD) All samples were one phase (Fig. 2), even that with 10 at% of Nd. The mean crystallite sizes were calculated from the Scherrer's formula and the sizes were about 52 nm for pure GGG and about 60 nm for GGG:Nd. There were not evident relationships between the crystallite sizes and amount of doping neodymium. We can observe a small shift of peaks increasing with increasing amount of dopant. The changes of the lattice constant in GGG:Nd samples are connected with increasing substitution gadolinium by neodymium (the size of neodymium ion is bigger than gadolinium ion) and peaks are shifted to the lower angle 2 theta, particularly for diffraction pattern no. 5 in figure 2 (GGG: 10 at% Nd). It is suggested that the changes of lattice constant between powdered crystals and nanopowders are connected with differences in defects and/or oxide stoichiometry in the structure.

High resolution scanning electron microscopy (HRSEM) In HRSEM images we can observe that morphology of all nanopowders pure GGG and doped by Nd is similar. Almost all powders have grains consisting of tubes with the shape of letters “U” or “V”. Only sample GGG: 0.2 at% Nd has more spherical grains (Fig. 3) and grains of the sample 10 at% Nd were more separated than these of lower concentration. The size of nanopowders grains increases with increasing Nd concentration.

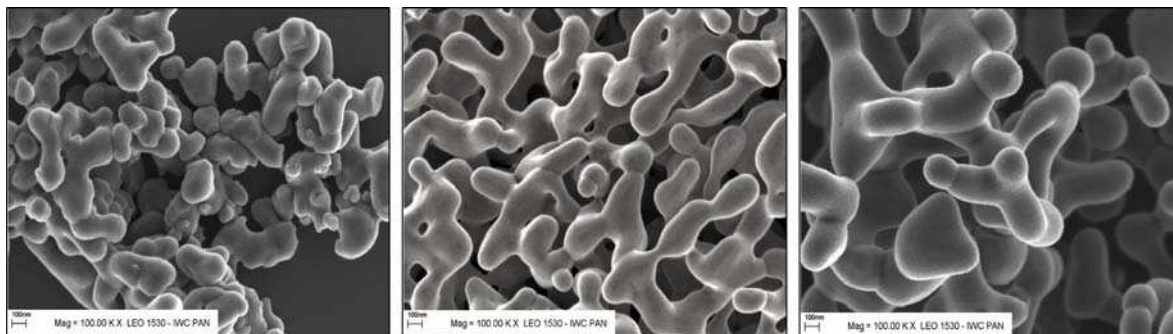


Fig. 3 HRSEM images of morphology of nanopowders: (A) $\text{Gd}_3\text{Ga}_5\text{O}_{12}$: 0.2 at% Nd^{3+} ; (B) $\text{Gd}_3\text{Ga}_5\text{O}_{12}$: 1 at% Nd^{3+} ; (C) $\text{Gd}_3\text{Ga}_5\text{O}_{12}$: 10 at% Nd^{3+} ; heated at 1100 °C and 12 h.

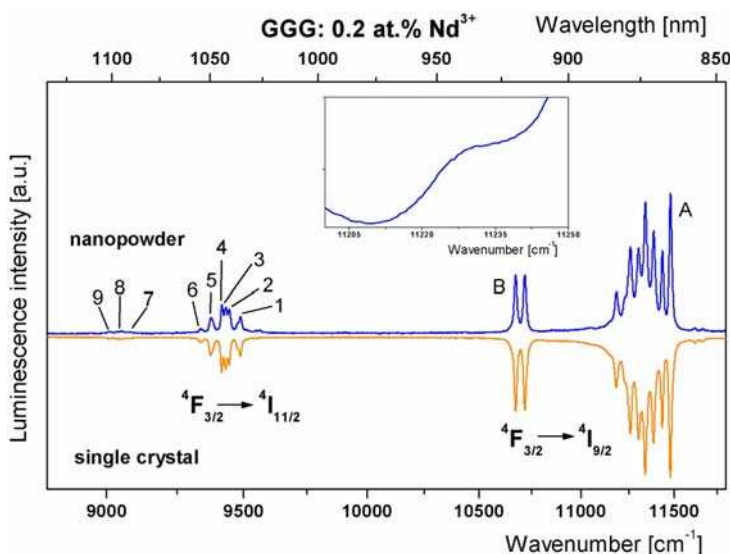


Fig. 4 Room temperature emission spectra for GGG: 0.2 at% Nd^{3+} nanopowder and GGG: 0.2 at% Nd single crystal excited at 806 nm. The spectrum recorded with a single crystal sample is inverted for the sake of clarity. As the insert is magnification of the spectra at 11205 – 11250 cm^{-1} . (Online color at www.crt-journal.org)

Spectroscopic features Figure 4 compares room temperature luminescence spectra related to the $^4\text{F}_{3/2} - ^4\text{I}_{9/2}$ and the $^4\text{F}_{3/2} - ^4\text{I}_{11/2}$ transitions of Nd^{3+} ions in a GGG nanopowder and in a single crystal grown by a micro-pulling-down technique. The spectra were excited at 806 nm by an AlGaAs laser diode, dispersed by a grating monochromator with 75 cm focal length and detected by a Hamamatsu R5108 photomultiplier. Luminescence band associated with the $^4\text{F}_{3/2} - ^4\text{I}_{9/2}$ transition consists of nine well-resolved narrow lines. There is also a shoulder clearly visible at about 11250 cm^{-1} implying that the actual number of band components is ten in agreement with expected number of transitions between two crystal field components of the $^4\text{F}_{3/2}$ initial manifold and five crystal field components of the terminal $^4\text{I}_{9/2}$ manifold.

The energy difference between the line peaking at 11480 cm^{-1} (denoted by A in the figure) and the line peaking at 10675 cm^{-1} (denoted by B in the figure) amounting to 805 cm^{-1} should correspond to the overall splitting of the ground $^4\text{I}_{9/2}$ state. The value derived is only slightly lower than 852 cm^{-1} reported for Nd:YAG. Predicted number of the luminescence band components for the $^4\text{F}_{3/2} - ^4\text{I}_{11/2}$ transition is twelve and at least nine lines can be discerned in the spectrum. Numbers 1-9 in the figure denotes them. It should be noticed that the

intensities of lines 7-9 in the spectrum shown is significantly lower than actual line intensities since the wavelengths of these transitions are close to the sensitivity limit of the photomultiplier. It can be seen in this figure that the spectra are very similar each other with respect to distribution of line intensities and line width. Numerical decomposition of spectra revealed that the band components recorded for single crystal samples are slightly much narrower. To discuss the peculiarities of spectra recorded we will refer to results of detailed analysis of the ${}^4F_{3/2}$ - ${}^4I_{11/2}$ emission band reported for a large Nd:GGG single crystal grown by the Czochralski method [27]. The spectra recorded at 4.2 K using site selective technique revealed existence of four non equivalent Nd sites in the GGG lattice, differing in energies of the crystal field levels. Comparison of low temperature selectively excited spectra to the room temperature spectrum recorded upon the broadband excitation has led to the conclusion that from four types of sites denoted by α , β , γ , δ solely the γ site contributes significantly to the room temperature emission spectrum. Based upon this assignment the laser line delivered by Nd:GGG single crystal at 1062.2 nm has been attributed to the transition between the highest crystal field component of the ${}^4F_{3/2}$ manifold and third crystal field component of the ${}^4I_{11/2}$ manifold of Nd^{3+} ions residing in the γ site.

The group of lines denoted by numbers 1-6 in our spectra shown in figure 4 resembles that reported for the Czochralski grown single crystal Nd:GGG with respect to distribution of the line intensities and the most prominent one (denoted by 4 in the figure) at 9416 cm^{-1} (1062.0 nm) agrees perfectly with the 1062.2 nm line observed for the single crystal. Also, spectral positions of remaining lines in the group denoted by 1-6 in this spectrum are consistent with the energies of the crystal field levels for the γ site in Nd:GGG single crystal found in ref. [17]. These results imply that the Nd^{3+} ions are located preferentially in the same type of sites in crystal lattices of Nd:GGG nanopowders and single crystals. In addition, it follows from the perfect agreement of emission wavelengths that the crystal lattice of GGG in nanopowders is not distorted in contrast to strong structural deterioration observed in Nd:GGG waveguide films fabricated by pulsed laser deposition [28].

Luminescence spectra recorded with our samples were the same with respect to line width and distribution of line intensities for all Nd^{3+} concentration studied up to 10 at% but the overall emission intensity decreased significantly with increasing Nd^{3+} content. This observation is consistent with the concentration dependence of luminescence decay curves shown in figure 5.

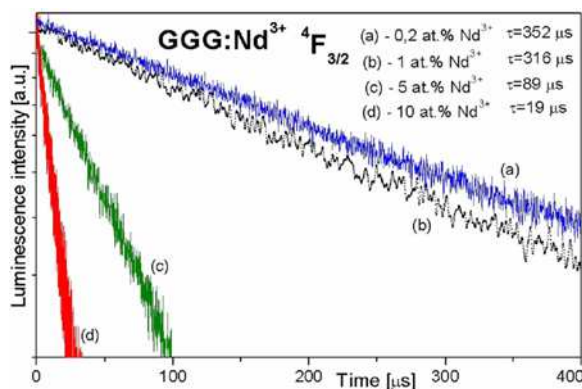


Fig. 5 Luminescence decay curves recorded at room temperature for GGG nanopowders with different Nd^{3+} concentrations: (a) 0.2 at%; (b) 1 at%; (c) 5 at%; (d) 10 at%. (Online color at www.crt-journal.org)

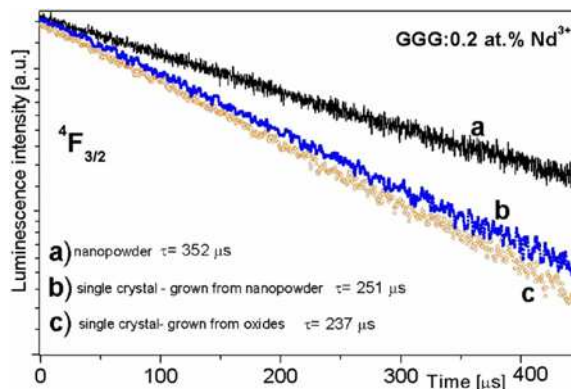


Fig. 6 Luminescence decay curves recorded at room temperature for GGG: 0.2 at% Nd^{3+} : (a) nanopowder; (b) single crystal grown from nanopowder; (c) single crystal grown from oxides. (Online color at www.crt-journal.org)

Decay curves in semilog plots are linear hence the decays follow single exponential time dependencies. For Nd^{3+} concentration of 0.2 at% the corresponding lifetime amounts to 352 μs , a value considerably higher than 249 μs reported for single crystal Nd:GGG [2].

Luminescence decay curves recorded imply important self-quenching of Nd^{3+} luminescence in GGG nanopowder. For Nd^{3+} concentration of 5 at% and for 10 at% the ${}^4F_{3/2}$ decay is governed mainly by

nonradiative relaxation. These findings are consistent with self-quenching effects observed in Nd:YAG single crystals. However, they disagree with small self-quenching of Nd emission in GGG that was claimed in paper [2] but not evidenced by experimental data. In figure 6 we compare luminescence decay curves acquired with GGG nanopowder, GGG single crystal grown by the μ -pulling down method starting from oxides and GGG single crystal grown by the μ -pulling down method starting from nanopowder, containing the same Nd³⁺ concentration.

It can be seen in this figure that decay times of the $^4F_{3/2}$ luminescence of neodymium in single crystal samples do not differ significantly whereas the luminescence decay in GGG:Nd nanopowder is considerably slower. This may be related to alteration of the crystal field within the surface layers in nanosized grains.

4 Conclusions

Nanopowders of GGG doped with neodymium up to 10 at% were prepared by using a sol-gel method modified in that the acetic acid was used as complexing agent for synthesis instead of citric acid. To compare the results of luminescence the crystals obtained by micro-pulling-down method were also investigated.

Luminescence spectra recorded with our samples were the same with respect to line width and distribution of line intensities for all Nd³⁺ concentration studied up to 10 at% but the overall emission intensity decreased significantly with increasing Nd³⁺ content as a consequence of strong self-quenching of luminescence. These findings are consistent with self-quenching effects observed in Nd:YAG single crystals. Nd³⁺ ions enter preferentially the same type of sites in nanopowders and single crystals of GGG. It can be concluded that the sol-gel method applied is able to provide nanopowders whose structural peculiarities are consistent with those encountered in single crystals. Also the single crystals prepared from the sol-gel nanopowders show good optical quality. Nanocrystalline powders open the door for new applications. They can be used as active face embedded in various matrix. Thus, the nanopowder GGG:Nd can be successfully applied in the future for optoelectronic and photovoltaic and as a starting material in the crystal growth process.

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